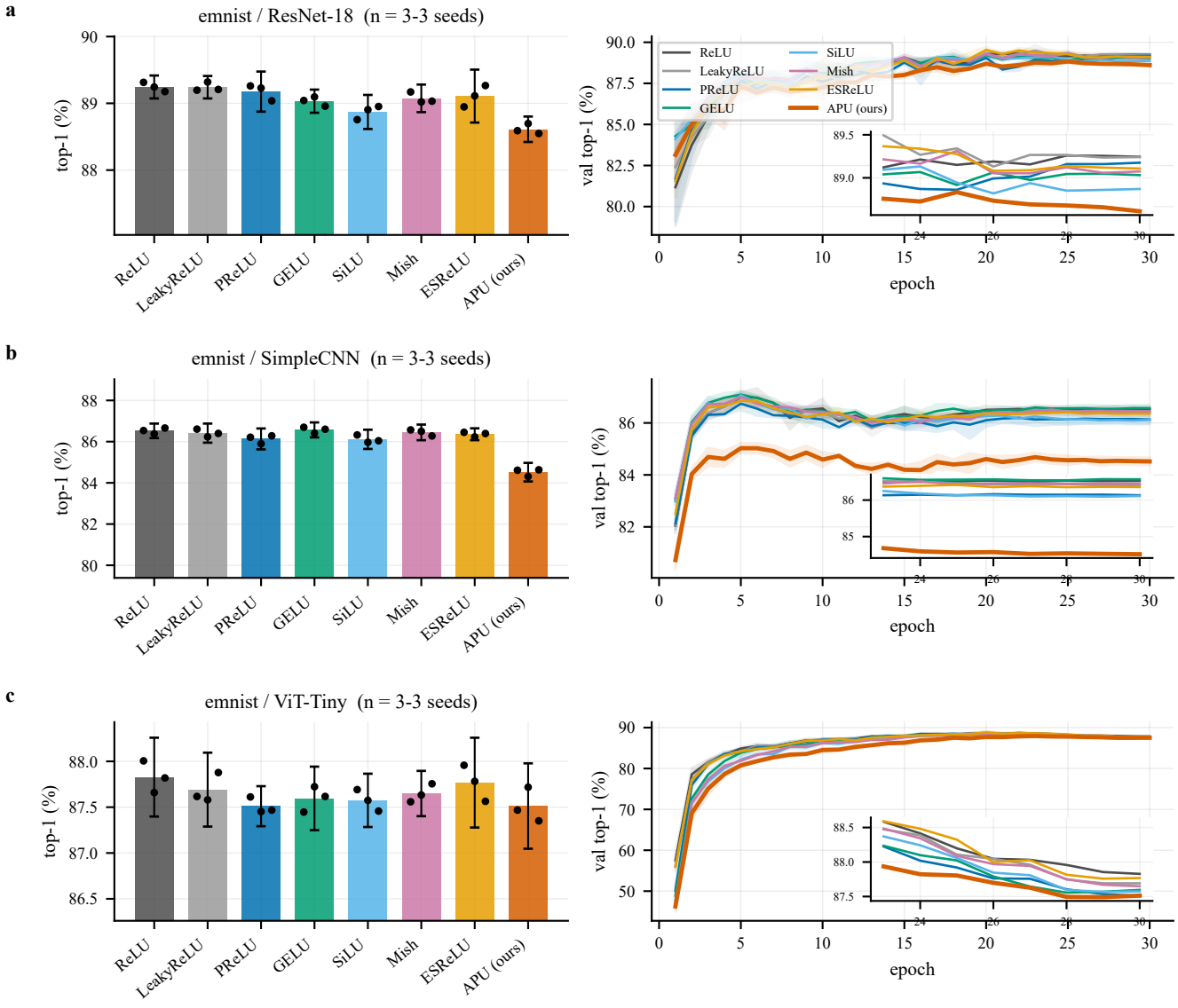

a
emnist / ResNet-18 (n = 3-3 seeds)
top-1 (%)
90
89
88
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)
val top-1 (%)
90.0
87.5
85.0
82.5
80.0
89.5
89.0
0
5
10
15
20
25
30
epoch
b
emnist / SimpleCNN (n = 3-3 seeds)
top-1 (%)
88
86
84
82
80
val top-1 (%)
86
84
82
86
85
epoch
c
emnist / ViT-Tiny (n = 3-3 seeds)
top-1 (%)
88.0
87.5
87.0
86.5
val top-1 (%)
90
80
70
60
50
88.5
88.0
87.5
epoch